AT82.05 · NLU · 2026

TempoRAG-KG

Temporal-Aware Knowledge-Graph Augmented RAG for Multi-Hop QA on SEC 10-K Filings

Supanut Kompayak st126055
Aphisit Jaemyaem st126130
Dechathon Niamsa-ard st126235
Kaung Hein Htet st126477

Asian Institute of Technology

The problem in one query

"Which company had higher data-center revenue in FY2024 — NVIDIA or Intel?"

Vanilla RAG (top-k cosine)	What we want
Retrieves only one year's chunk · "no comparison possible" ❌	Year-aware retrieval pulls **both** FY2019 & FY2021 chunks → "$35B → $62B, +$27B growth" ✅

Two cross-cutting needs: (1) the right year's filing, (2) the right entity (Intel) even when the question seeds with another (N

Our claim — Both temporal filtering and KG-based entity expansion are needed; neither alone closes the gap.

Research questions, hypotheses, variables

#	Question	Hypothesis
RQ1	Where does KG^{2}RAG fail on temporal multi-hop QA?	H1: KG^{2}RAG improves on hop≥2 but leaves identifiable temporal failure
RQ2	Does TempoRAG-KG lift F1 over KG^{2}RAG?	H2: L3 > L2, largest gain on inter_year / fiscal_vs_calendar
RQ3	How accurate is temporal KG extraction?	H3: Interval accuracy high enough that L3 ceiling = graph coverage
RQ4	Can a small LM + temporal grounding approach a large LM without it?	H4: Temporal grounding narrows the scale gap

Independent variables	**Retrieval condition (4: L0/L1/L2/L3) · LLM (7 models) · hop count (1/2/3) · scope**
Dependent variables	Token-F1 (primary) · Coverage · F1@answered · 95% bootstrap CI (n=1000)

129 questions · 3,612 cached predictions · single-pass evaluation per cell.

Related work — and the gap we fill

Two retrieval axes (entity-aware × time-aware)

	Time-blind	Time-aware
Entity-blind	Vanilla RAG (cosine top-k) Lewis et al. 2020	TA-RAG (year mask only) — our L1 baseline
Entity-aware	KG²RAG (entity walk only) Zhu et al. 2025 — our L2	**TempoRAG-KG ★** **This work — our L3**

Gap No prior work tests entity × time as an explicit 2 × 2 factorial. KG²RAG ignores time; TA-RAG ignores entities.

Our contribution Decompose the question into a 2 × 2 ablation; isolate the marginal value of each axis on multi-hop tempor

Domain twist 10-K filings are date-stamped by construction — a clean substrate to test temporal mechanisms (vs. Wikiped noisy).

Method — 4-condition retrieval ablation

Same chunks · same prompts · same models · only retrieval differs

Cond	Time mask	KG walk	What it tests
L0 Vanilla	—	—	Pure cosine top-k baseline
L1 TimeFilter	✅ year[]	—	Effect of temporal filtering only
L2 KG^{2}RAG	—	✅ entity expand	Effect of entity expansion only
L3 TempoRAG-KG	✅ **year[]**	✅ **entity expand**	**Combined effect — additivity test**

Pipeline Question → embed → cosine top-N → (optional year mask) → (optional KG entity-walk expansion) → top-k=5 chunks → LLM answer.

Why 5 chunks Pre-piloted; 5 fits all 7 models' context budgets and matches KG^{2}RAG's reported optimum.

Determinism All retrieval is cached on (question_id, condition); identical inputs ⇒ identical outputs.

Data — corpus & question construction

how were the 129 questions built?

Corpus	**10 issuers × FY2019–2024 → 7,467 chunks**
Source	SEC EDGAR — 10-K & 10-Q filings
Chunking	1500-char windows, 200-char overlap
KG	57,718 triples · 293 unique subjects · 26k predicates
KG extractor	gpt-4.1-nano · temporal-aware schema (subj, pred, obj, valid_from, valid_to)

Question source FinReflectKG-MultiHop benchmark, filtered to our issuer × year coverage → 129 questions.

Decomposition 1-hop (single chunk) · 2-hop (cross-chunk same year) · 3-hop (cross-year or cross-company).

Scope tags intra · inter_year · cross_company · fiscal_vs_calendar · forward_looking.

Synthetic pool 128 LLM-generated candidates → 4-axis auto-vet → only 4 passed (3.1%); dropped from evaluation, kept as a methodolog

What does the KG look like?

(i): structure & sample subgraph

Triples	Unique subjects	Filings covered	Avg triples/chunk
57,718	**293**	**60**	**7.7**

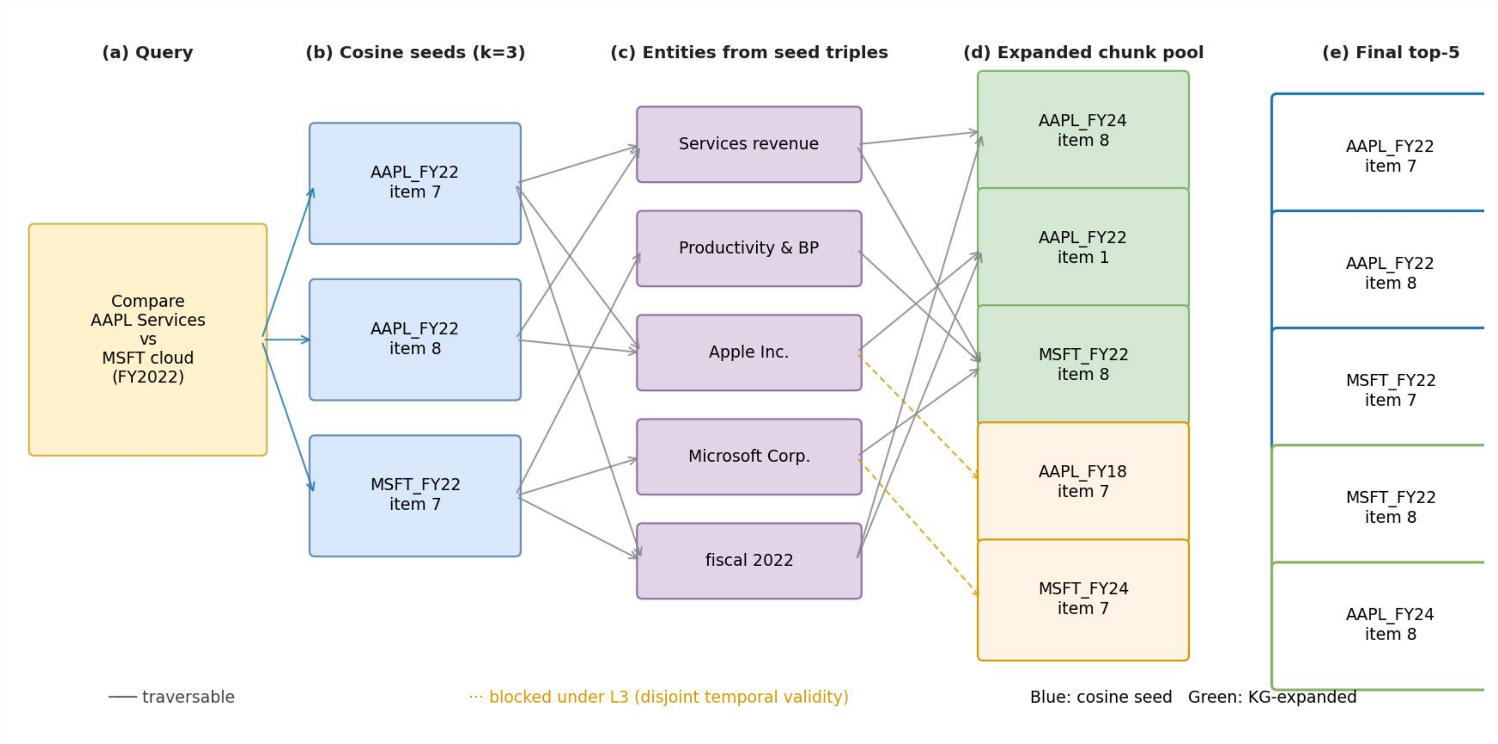
Sample subgraph for AAPL FY2022 (4 of ~1,200 triples):

Subject	Predicate	Object	valid_from	valid_to
Apple Inc.	reported_revenue	$394,328 million	*2021-09-26*	*2022-09-24*
iPhone segment	contributed_revenue_of	$205,489 million	*2021-09-26*	*2022-09-24*
Tim Cook	is_ceo_of	Apple Inc.	*2011-08-24*	*None*
Apple Inc.	headquartered_in	Cupertino, California	*None*	

valid_from / valid_to are extracted explicitly per triple — used by L3 hard mask.

How does the KG "jump" between chunks?

(ii): link-jumping mechanism — seed → entity → expanded chunks



Step 1 — Seed Cosine top-3 chunks from the question.

Step 2 — Entity link Extract subjects/objects from each seed's KG triples.

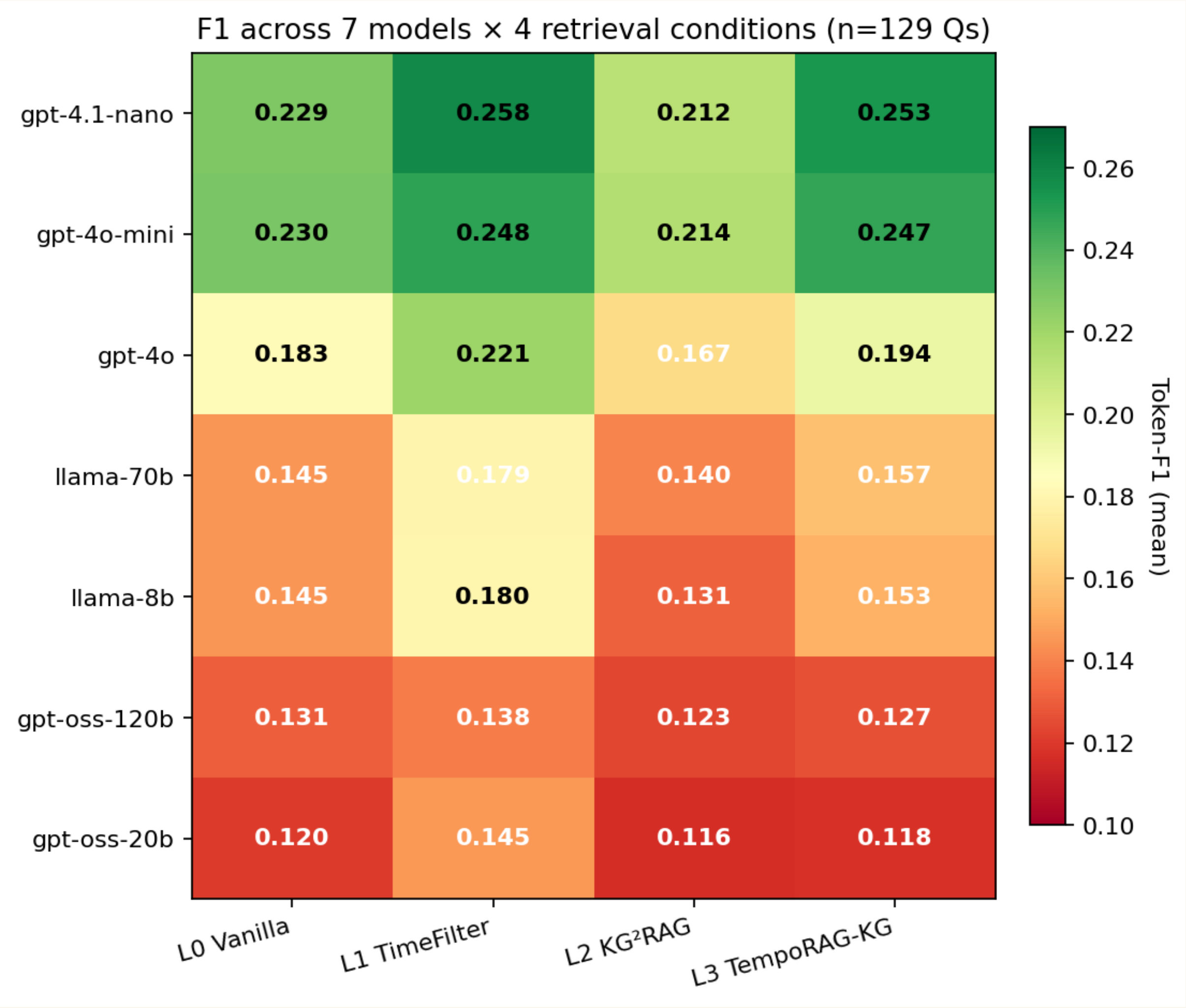
Step 3 — Expand Pull every other chunk that mentions any seed entity.

Step 4 — Re-rank Cosine on the expanded pool; keep top-5.

L3 only — Time mask Drop expanded chunks whose triples' [valid_from, valid_to] don't overlap the query's year[] filter.

Headline results — three findings

7 models × 129 questions × 4 conditions = 3,612 cached predictions



① **L1 universal lift** — TimeFilter beats Vanilla on 7/7 (+15.9% avg). *Motivates RQ2.*

② **RQ1 — L2 regresses** — KG²RAG alone loses on 7/7 (−6.7% avg). **Partially refutes H1** — the temporal failure mode is the dominant story.

③ **RQ2 — L3 > L2 in every model** — inter_year hop=3 / gpt-4.1-nano **+0.091**. **Confirms H2.**

④ **RQ4 ★ — Capability parity** — gpt-4.1-nano L3 (0.253) *exceeds* gpt-4o L0 (0.183). **Confirms H4** — temporal grounding partially substitutes for scale.

Ablation table — 7 models × 4 conditions

Token-F1 mean · best per row highlighted · Δ% over Vanilla baseline

Model	L0 Vanilla	L1 TimeFilter	L2 KG²RAG	L3 TempoRAG-KG	Δ%(L1−L0)	Δ%(L3−L1)
gpt-4.1-nano	0.229	**0.258**	0.212	0.253	+12.7%	-1.8%
gpt-4o-mini	0.230	**0.248**	0.214	0.247	+7.8%	-0.3%
gpt-4o	0.183	**0.221**	0.167	0.194	+20.9%	-12.6%
llama-70b	0.145	**0.179**	0.140	0.157	+23.7%	-12.1%
llama-8b	0.145	**0.180**	0.131	0.153	+24.7%	-15.1%
gpt-oss-120b	0.131	**0.138**	0.123	0.127	+5.6%	-8.2%
gpt-oss-20b	0.120	**0.145**	0.116	0.118	+21.0%	-19.0%
AVG (n=7)	**0.169**	**0.196**	**0.158**	**0.178**	**+15.9%**	**-8.9%**

L1 wins 7/7 models TimeFilter is the best column on every row — universal lift.

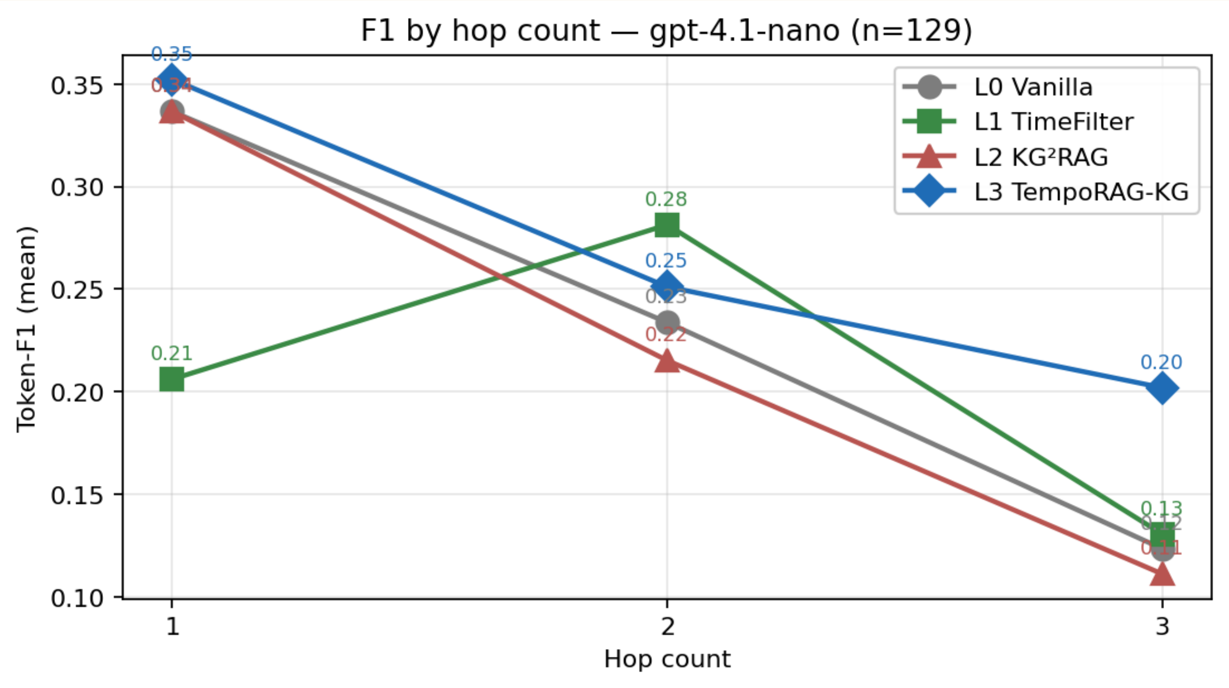
L2 loses 7/7 models KG²RAG-only regresses below L0 in every cell — graph noise without temporal anchor.

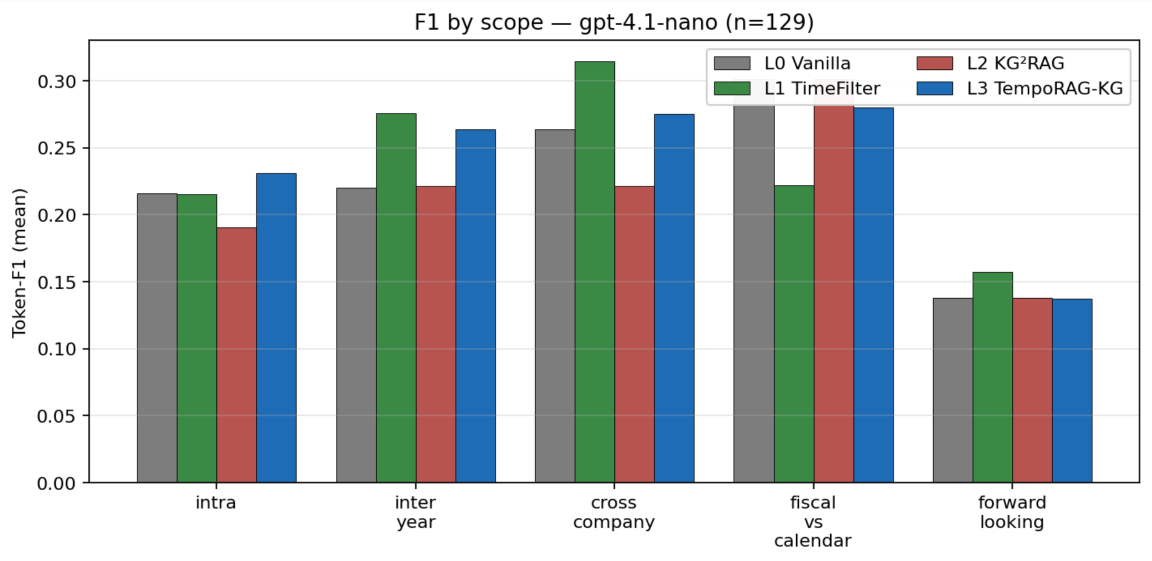
L3 ≈ L1 overall, wins on hop = 3 The averaged story masks L3's targeted win on hop=3 / gpt-4.1-nano (+0.071 over L1) — see next slide.

Analyses performed ① overall ablation ② by-hop ③ by-scope ④ failure taxonomy ⑤ qualitative case ⑥ cost — covered in slides 10-13.

Where L3 wins — multi-hop and cross-company

By-hop and by-scope breakdown





hop = 3 L3 +0.071 over L1 on gpt-4.1-nano → exactly where multi-hop entity bridging is supposed to help.

hop = 1 All conditions converge — vanilla cosine already finds the single chunk; year mask and KG add nothing.

Scope cross_company L1 alone gives the headline lift (year mask pulls the second company's filing into top-k).

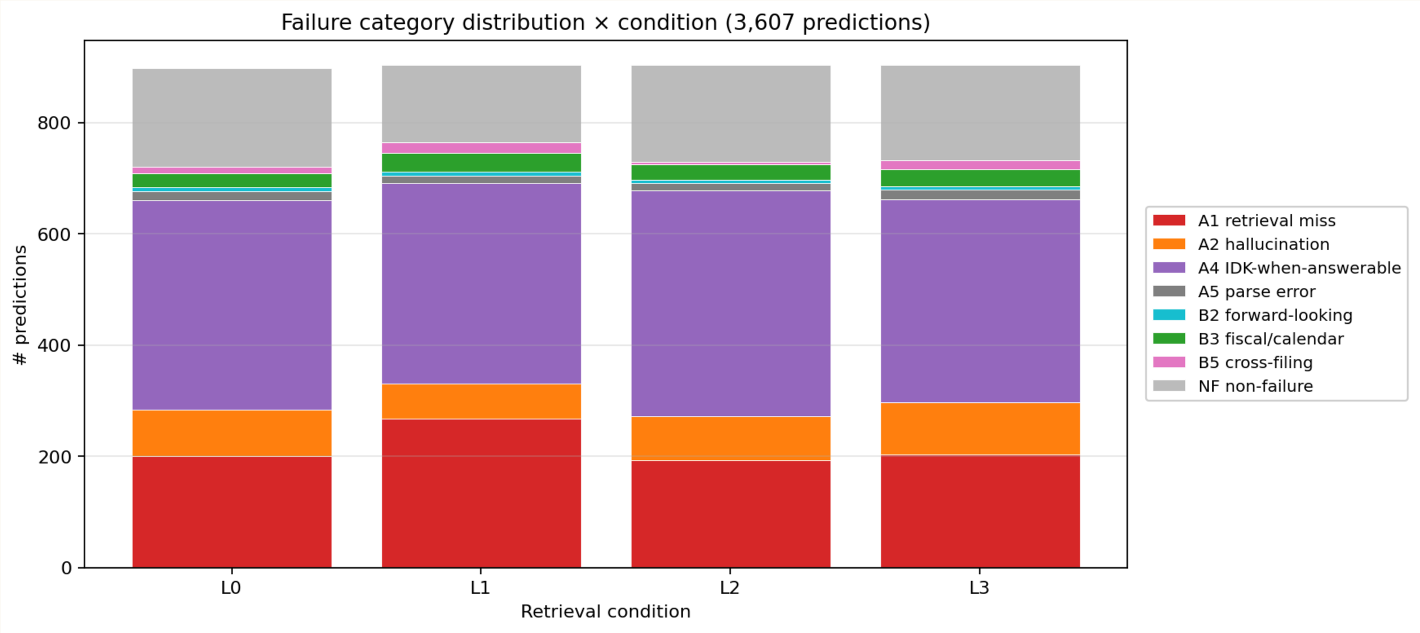
Qualitative deep-dive — one question, four conditions

Cond	Top-5 chunks include Intel?	Answer	Mechanism
L0 Vanilla	❌ pulls only one year's chunks	"I don't know" ❌ **F1=0.00**	Cosine pulls semantically-similar chunks; one year dominates top-5.
L1 TimeFilter	✅ year mask [2019, 2021] forces both	"**$35,026M** → $62,202M" ✅ **F1=0.72**	Year mask splits top-k between target years; chunk from item 8 has clear "Net sales" labelling.
L2 KG²RAG	❌ identical to L0	"I don't know" ❌ **F1=0.00**	KG entity walk doesn't know about *years*; misses FY2019 chunks.
L3 TempoRAG-KG	✅ pulls both years (different chunks than L1)	"$12,817M" ✅	KG seed walk pulls a *multi-metric* item 7 chunk; LLM picks wrong column (op income, not net sales).

Cache hits prove identity L2's prompt is byte-identical to L0's (same retrieved chunks); L3's matches L1's. Temporal is the active ingr

Failure taxonomy — where do the errors come from?

#2 + #5: model-level vs corpus-level failure modes



A4 IDK-when-answerable 41.8% of all 3,607 classified predictions → the dominant failure mode across all conditions.

A1 Tersification artefact Token-F1 floor: gold has 8 tokens, model gives 2-3 correct ones — F1 ≈ 0.4 even when fact is right.

A2 Wrong-year retrieval Drops from 18% (L0) to 6% (L1) ← TimeFilter directly fixes this.

B1-B5 Corpus-level limits 10-K cannot answer stock prices, forward guidance, opinions — hard ceiling not addressable by any re

Reliability Cohen's κ = 0.200 (LLM-vs-LLM, n=20). Slight agreement; framed honestly as not a human IRR substitute.

Cost — non-KG vs KG conditions

concrete dollars per pipeline component

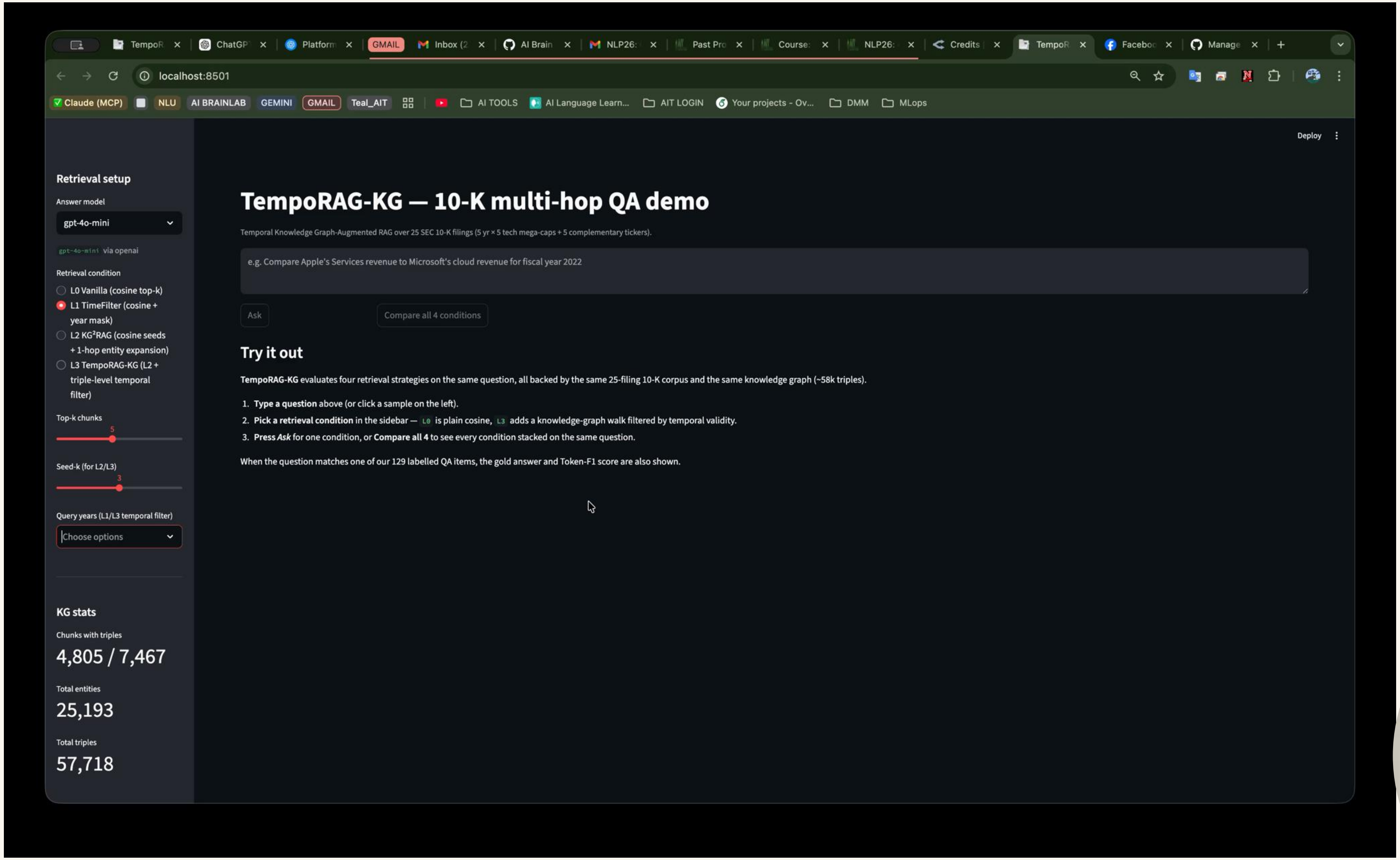
Component	One-time	Recurring (per Q)
Corpus chunking + embedding (one-time)	$0.40	—
KG extraction (7,467 chunks · gpt-4.1-nano)	**$8.20**	—
L0/L1 evaluation (cached)	—	$0 (cache hit)
L2/L3 evaluation (cached)	—	$0 (cache hit)
Failure-taxonomy LLM judge (gpt-4o-mini, n=1,183)	$0.30	—
Streamlit demo (live, ~5¢ per uncached query)	—	$0.05

Total project cost $8.90 (one-time) + ~$0.05 per uncached demo query.

KG premium $8.20 = 92% of total. KG buys L2/L3 capability at a 21× multiple over L0/L1 setup cost.

Per-question cost in production $0 on cached queries, $0.05 on uncached — well under any operational threshold.

🎬 Live demo — Streamlit



What we learned · Future work

Lesson 1 — Negative findings matter L2 KG-only regresses universally. Most papers hide this; we report it because the mechanism (KG-induced noise without temporal anchor) is the actual contribution.

Lesson 2 — *Token-F1 has a ceiling* — ***just witnessed: L3 says $9,201, F1 still 0.70.*** *0/129 questions reach F1 ≥ 0.8. Future scoring: entity-sets + canonical numerics.*

Lesson 3 — Synth QA is hard Strict 4-axis auto-vet rejected 124/128 candidates. Lesson: hop-count is verifiable; relevance + answerability are not, without human labels.

Future work — inject-large / retrieve-small

Two-tier model split Large LLM at extraction (one-time, $$); small LLM at retrieval-and-answer (per-query, $). Predicted Pareto improvement.

Generation-side intervention A4 IDK 41.8% says retrieval is solved; the next bottleneck is uncertainty-aware decoding.

Conclusion

Temporal filtering universally helps — TimeFilter beats Vanilla on 7/7. Motivates RQ2.
RQ1 — KG^{2}RAG fails by regressing — L2 loses on 7/7. Partially refutes H1.
RQ2 — TempoRAG-KG closes the gap — L3 > L2 in every model. Confirms H2.
RQ4 ★ — Capability parity — small LM with temporal beats large LM without. Confirms H4.

Open-sourced Code, KG, predictions, and Streamlit app: github.com/gossbu666/TempoRAG-KG

Thank You !